

P1g: the hazard induction for $Z_N^\pm$ and the ω -conditions of Theorem C at every root of unity of the middle arc

research programme P1, seventh atom, 2026-09-24. Reviewed adversarially (Reviewer AC); all gaps fixed, all
certificates rerun.

Notation of P1d.lemma.tex, P1e.lemma.tex and P1f/P1f.lemma.tex (compiled numbering: P1d Theorem 2 Gauss factorisation, Theorem 5 Fresnel, Lemma 6 dec, Lemma 8 lap, Theorem 9 corrected lemma; P1e Lemmas Zk, R, A, H, B, B', O and Theorem M; P1f Theorem C). Labels: PROVED (complete proof here or in the cited notes); *computer-assisted* (reduced to finitely many Arb ball-arithmetic inequalities checked by the named script, which fails loudly); VERIFIED (numerics, not a proof); HEURISTIC; OPEN.

Outcome.

- (a) **Theorem 14 (PROVED, computer-assisted).** For every $N \geq 3$ and every a with $\gcd(a, N) = 1$, $\frac{a}{N} \in [\frac{1}{5}, \frac{4}{5}]$, $\frac{a}{N} \neq \frac{1}{2}$:
- $|Z_N^+|, |Z_N^-| \geq 0.02 e^{-0.005N}$; in particular $Z_N^+ \neq 0$;
 - $|\omega| \leq 7.96$, and every one of the six normalised transfer factors G_i of Definition 1 has modulus ≥ 0.0499 . So ω avoids the six values of P1f Theorem C(iii).

The proof adapts the hazard induction of P1e Theorem M to $Z^\pm$ (Lemmas 6–9). It carries a quantitative ω -margin along the chain of deepest hazards (Lemma 9, Lemma 3).

- (b) **Corollary 15 (PROVED, computer-assisted; standing Theorems thm:model, thm:RJ as in P1f).** For every $N \geq 16$ and every $\zeta = e(a/N)$ with $\frac{1}{5} \leq \frac{a}{N} \leq \frac{4}{5}$, the poles of U and V accumulate at $\pm\sqrt{\zeta}$ and $\pm\sqrt{\bar{\zeta}}$. Consequently the poles of U and of V accumulate at every point of the arcs $\arg x \in [\frac{\pi}{5}, \frac{4\pi}{5}] \cup [\frac{6\pi}{5}, \frac{9\pi}{5}]$. This closes item 1 of §6 of P1f (“density of the poles of U and V on the arcs: OPEN”).
- (c) **What made it work.** $Z_N^\pm$ has the same single-saddle structure as Z_N , with the weights γ_k shifted to $\gamma_{k\pm 1}$ and the amplitude multiplied by $e^{\mp\pi it}$ (Lemma 6). So the renormalisation of P1e Lemma R carries over unchanged: near a seed j/n , $Z_N^\pm$ inherits $Z_n^\pm(j/n)$, and ω_N inherits $\omega_n(j/n)$ up to an exponentially small relative error (Lemma 9). A further point is new. P1e’s direct Lemma A (“ $Z = 1 + \text{small}$ ”) is too crude to control ω . It is replaced by an exact low-order expansion with a rigorous majorant tail (Proposition 12). That expansion is a continuous function of $x = a/N$, uniform in $N \geq 151$, and it covers the whole complement of the hazards and of shrunken low-seed discs. The covering has 12,205 intervals and takes about 10 s.

1 Objects

Let $\zeta = e(x)$, $x = a/N$, $u_0 = \lambda^2/c$, $e^{\Phi(Y)} = \sum_n \psi_n Y^n$ as in P1d Theorem 2. For $s \in \{0, +1, -1\}$ put

$$Z_N^s := \sum_{n \geq 0} \psi_n \zeta^{-n^2 - sn} \quad (Z_N^0 = Z_N, \quad Z_N^{\pm 1} = Z_N^{\pm} \text{ of P1f}), \quad \omega := \frac{Z_N^-}{u_0 Z_N^+}, \quad t_1^* := -\frac{1}{2}(1 + \omega).$$

Definition 1 (normalised transfer factors). With $X := e(x/2)$ (so $\{X, -X\} = \{\pm\sqrt{\zeta}\}$) and $t = t_1^*$:

$$G_1 = \frac{1 - \zeta^2}{\zeta} - t, \quad G_2 = \frac{1 - \zeta}{\zeta} - t, \quad G_3^{\pm} = t + \frac{1 \pm X}{\pm 2X}, \quad G_4^{\pm} = t(1 \mp X + \zeta) + \frac{1 + \zeta}{\pm 2X}.$$

Each is $p_i(x)t + q_i(x)$ with $p_i \in \{-1, 1, 1 \mp X + \zeta\}$, so $|p_i| \leq 3$.

Lemma 2 (PROVED). For $\zeta \neq \pm 1$: all six $G_i \neq 0 \iff \omega \notin \{-1 - \frac{2(1-\zeta^2)}{\zeta}, -1 - \frac{2(1-\zeta)}{\zeta}\} \cup \{\frac{1}{y}, \frac{1+\zeta}{y(1-y+\zeta)} - 1 : y = \pm\sqrt{\zeta}\}$. This is the hypothesis of P1f Theorem C(iii). When $1 - y + \zeta = 0$ (this happens only for $\zeta = e(\pm\frac{1}{3})$, $y = e(\pm\frac{1}{6})$) the corresponding excluded value is ∞ and $G_4 = (1 + \zeta)/(2y) \neq 0$ automatically.

Proof. $1 - \mathfrak{g}_U t_1^* = 0 \iff t_1^* = (1 - \zeta^2)/\zeta \iff \omega = -1 - 2(1 - \zeta^2)/\zeta$, and similarly for $\mathfrak{g}_V = \zeta/(1 - \zeta)$. $G_3 = 0 \iff 1 + \omega = 1 + 1/y$. $G_4 = 0 \iff (1 + \omega)(1 - y + \zeta) = (1 + \zeta)/y$. $\square$

The normalisation removes the poles of the P1f factors: $\mathfrak{g}_U$ blows up at $\zeta = -1$, and the factor Π_q blows up at $1 - y + \zeta = 0$. The coefficients are bounded and Lipschitz:

Lemma 3 (Lip; PROVED). For real x, x' and $t, t' \in \mathbb{C}$, and each i :

$$|G_i(x, t) - G_i(x', t')| \leq 3|t - t'| + |x - x'| (3\pi|t'| + 4\pi).$$

Proof. $|p_i| \leq 3$. $|\frac{d}{dx} p_i| \leq 3\pi$, since $X' = \pi i X$ and $\zeta' = 2\pi i \zeta$. The q_i are $\zeta^{-1} - \zeta$, $\zeta^{-1} - 1$, $\frac{1}{2} \pm \frac{1}{2X}$ and $\pm\frac{1}{2}(X^{-1} + X)$, with derivatives of modulus $\leq 4\pi, 2\pi, \frac{\pi}{2}, \pi$. $\square$

Lemma 4 (conjugation; PROVED). $Z_N^s(\frac{N-a}{N}) = \overline{Z_N^s(\frac{a}{N})}$ for each s . Hence $\omega(\frac{N-a}{N}) = \overline{\omega(\frac{a}{N})}$, and the multiset $\{|G_i|\}$ is the same at $\frac{a}{N}$ and $\frac{N-a}{N}$.

Proof. As P1d Lemma 4. Every ingredient is conjugated, and $\zeta^{-n^2 - sn} \mapsto \bar{\zeta}^{-n^2 - sn}$. The G_i have real rational coefficients in ζ, X , and $X \mapsto \bar{X}$ is a root of $\bar{\zeta}$. $\square$

So it suffices to treat $x \in [\frac{1}{5}, \frac{1}{2})$.

2 The single-saddle structure of $Z^{\pm}$

Lemma 5 (Zpm, finite formula; PROVED). At a reduced p/q (level q , Φ_q excludes $q \mid n$): $Z_q^s(p/q) = \sum_{k \bmod q} \gamma_k^s e^{\Phi_q(\zeta_0^k)}$, where $\gamma_k^s = \frac{1}{q} \sum_{\nu \bmod q} \zeta_0^{-\nu^2 - s\nu - k\nu} = \gamma_{k+s}$.

Proof. $n \mapsto \zeta_0^{-n^2 - sn}$ is q -periodic; finite Fourier inversion and absolute convergence exactly as P1e Lemma Zk. $\square$

Lemma 6 ($F^{\pm}$, twisted Fresnel representation; PROVED). In the setting of P1d Theorem 5 (seed p/q , $x = p/q + d$, $d > 0$, contour Γ):

$$Z_N^s = e(\frac{d}{4}) \frac{e^{-i\pi/4}}{\sqrt{2d}} \sum_{k \bmod q} \gamma_k^s J_k^s, \quad J_k^s = \int_{\Gamma} e^{\pi i t^2 / (2d)} e^{-s\pi i t} e^{\Phi(\zeta_0^k e(-t))} dt.$$

Proof. $\zeta^{-n^2-sn} = \zeta_0^{-n^2-sn} e(-d(n + \frac{s}{2})^2) e(\frac{d}{4})$. Expand the first factor by Lemma 5. For $m = n + \frac{s}{2}$ the Fresnel identity $\int_{\Gamma} e^{\pi i t^2/(2d)} e(-mt) dt = \sqrt{2d} e^{i\pi/4} e(-dm^2)$ holds for every real m : complete the square and shift the contour to $\Gamma - 2dm$, as in P1d. Also $e(-mt) = e(-nt) e^{-s\pi i t}$. Fubini holds as in P1d. On Γ , $\sum |\psi_n| |e(-nt)| \leq \sum |\psi_n| e^{2\pi n\eta} < \infty$. The factor $|e^{-s\pi i t}| \leq e^{\pi|t|}$ is dominated by the Gaussian in both end sectors. $\square$

So Z^s is P1d's integral with amplitude $b_k^s(t) = e^{B_k(t)} e^{-s\pi i t}$ (P1d Lemma 6 is unchanged), and the main term is

$$\text{Main}^s = e(\frac{d}{4}) e^{i(\alpha-\pi/4)} \sqrt{\frac{\pi}{2A}} e^{F(t_0)/d} S^s, \quad S^s = \sum_k \gamma_k^s e^{B_k(t_0)-s\pi i t_0}.$$

The prefactor of S^s does not depend on s .

Lemma 7 (Lap $^\pm$; PROVED). *P1d Lemma 8, and its sharpening P1e Lemma O, hold for Z_N^s/Main^s with the following changes:*

1. the weights are γ_k^s ;
2. on the core segment, B_m, B_d, K_b are the bounds for $b_k^s, (b_k^s)', (b_k^s)''/2$, where $(b^s)' = (B'_k - s\pi i)b^s$;
3. each outer piece of Γ carries the extra factor $\sup |e^{-s\pi i t}| = \sup e^{s\pi \text{Im } t}$ over that piece;
4. on the unbounded part of the lower ray ($|v| \geq V, t = t_0 + e^{i\alpha}v$), the growth $e^{\pi|v|\sin\alpha}$ is absorbed by halving the Gaussian constant: $c \rightarrow c/2$, valid as soon as $V \geq 2\pi d/c$. The factor $e^{\pi|\text{Im } t_0|}$ is kept.

Proof. The proofs of P1d Lemma 8 and P1e Lemma O use only analyticity of the amplitude and the bounds listed. The core expansion $e^{\psi/d}b - b_0 = b_1v + b_0c_3v^3/d + R$ is unchanged, with $b = b^s$. Each outer integral $\int |e^{(F-F_0)/d}| |b^s| |dt|$ is bounded by (sup of the twist on the piece) $\times$ (the old bound). On the lower tail, $\pi|v| \leq \frac{c}{2d}v^2$ for $|v| \geq 2\pi d/c$. So $\int_V^\infty e^{-cv^2/d+\pi v} dv \leq \int_V^\infty e^{-cv^2/(2d)} dv$. $\square$

Implemented in `P1g.cert.py` (low seeds; Lemma O form; the twist is enclosed exactly on the core and at t_0). In the crude uniform form of P1e Lemma B it is implemented in `P1g_lemmaB.py`. There:

- $\beta \rightarrow \beta e^{\pi\eta}$ on the core, at t_0 and on the horizontal part (valid because $R_0 + v_0 < \eta$);
- $P_1 \rightarrow P_1 + \pi$;
- $r_4 \rightarrow \sqrt{2A/A'} \text{erfc}(v_0\sqrt{A'/(2d)}) \beta e^{\pi R_0} e^{\delta\eta}$, with the side condition $v_0 \geq 2\pi d/A'$ checked;
- $r_5 \rightarrow r_5 e^{\pi\eta}$.

Lemma 8 (R $^\pm$; PROVED). *With $T^s := (1 - w_q)^{-1/(2q)} Z_q^s(p/q)$ and $\text{env}^s := \sum_k |\gamma_k^s| e^{X_k}$:*

$$\left| e^{s\pi i t_0} S^s - T^s \right| \leq \text{env}^s (e^{\delta_R} - 1),$$

with δ_R exactly as in P1e Lemma R. In particular $|S^s| \geq e^{-\pi|\text{Im } t_0|} (1 - R^s) |T^s|$, where $R^s := \text{env}^s (e^{\delta_R} - 1) / |T^s|$.

Proof. $e^{s\pi i t_0} S^s = \sum_k \gamma_k^s e^{B_k(t_0)}$. P1e's proof bounds $|B_k(t_0) - X_k| \leq \delta_R$ for each k , independently of the weights. Lemma 5 identifies $\sum_k \gamma_k^s e^{X_k} = T^s$. $\square$

Lemma 9 (I: inheritance of ω ; PROVED). *Suppose that, at the seed p/q and the point $x = a/N$, for $s = \pm 1$:*

- $|Z_N^s/\text{Main}^s - 1| \leq E < 1$;
- $R^s \leq R < 1$;
- $\epsilon_u = |\tilde{u}/u_q - 1| < 1$, with $\tilde{u} = u_0 e(-t_0)$ as in P1e Lemma R.

Then $\omega_N = \omega_q(p/q) \cdot \frac{u_q}{\tilde{u}} \cdot Q$, with $|Q - 1| \leq \left(\frac{(1+R)(1+E)}{(1-R)(1-E)}\right)^2 - 1$. Hence

$$\left| \frac{\omega_N}{\omega_q(p/q)} - 1 \right| \leq \rho := \left(1 + \frac{\epsilon_u}{1 - \epsilon_u}\right) \left(\frac{(1+R)(1+E)}{(1-R)(1-E)}\right)^2 - 1.$$

With Lemma 3, every $|G_i|$ at (x, ω_N) is at least $\min_i |G_i|$ at $(p/q, \omega_q)$ minus

$$\sigma := \frac{3}{2}|\omega_q|\rho + |d|\left(\frac{3\pi}{2}(1 + |\omega_q|) + 4\pi\right).$$

Proof. Write $Z_N^s = \text{Main}^s(1 + \theta_s)$ and $e^{s\pi i t_0} S^s = T^s(1 + \varepsilon_s)$, with $|\theta_s| \leq E$ and $|\varepsilon_s| \leq R$. The common prefactor cancels in Z_N^-/Z_N^+ , and $S^-/S^+ = e(t_0) \frac{T^-}{T^+} \frac{1+\varepsilon_-}{1+\varepsilon_+}$ with $T^-/T^+ = Z_q^-/Z_q^+$. So $\omega_N = \frac{Z_q^-}{u_0 e(-t_0) Z_q^+} Q = \omega_q \frac{u_q}{\tilde{u}} Q$, where $Q = \frac{(1+\varepsilon_-)(1+\theta_-)}{(1+\varepsilon_+)(1+\theta_+)}$. The loss σ follows from Lemma 3 with $|t - t'| = |\omega_N - \omega_q|/2 \leq |\omega_q|\rho/2$ and $|t'| \leq (1 + |\omega_q|)/2$. $\square$

VERIFIED (P1g_inherit_check.py, mpmath): along $x_N = j/n - 1/(nN)$ at $1/3, 2/5, 3/7, 4/11$, $|\omega_N - \omega_n(j/n)|$ is $0.07, 0.025, 0.010$ at $|d| = 5.5, 1.8, 0.74 \cdot 10^{-3}$ ($1/3$), i.e. $O(d)$. The ratios $Z_N^\pm/Z_N$ converge to $Z_n^\pm/Z_n$ as well.

3 The certificates

Proposition 10 (S: seeds and base; PROVED, computer-assisted). *For all 4116 reduced j/n with $3 \leq n \leq 150$, $n \leq 5j \leq 4n$, $j/n \neq \frac{1}{2}$ (P1g_seeds.py, blocks 3–70, 71–105, 106–130, 131–150, all SEEDS PASSED):*

$$|Z_n^\pm(j/n)| \geq 0.6067, \quad \min_i |G_i| \geq 0.2331, \quad |\omega_n| \leq 4.2352.$$

The files `seeds_*.txt` give, per seed and in directed rounding: $|Z_n^s|$, $\text{env}^\pm$, $|\omega_n|$ and the margin.

The enclosures use Lemma 5 with the rigorous Φ -tail of P1f.

Proposition 11 (L: low seeds; PROVED, computer-assisted). *For each of the 18 low seeds of P1e (side $d > 0$; the side $d < 0$ of p/q is the side $d > 0$ of $(q-p)/q$), P1e's parameters $r, \eta, V_0, n_2 = 60$ and $\kappa = 3$, and 8 geometric pieces, P1g_lowseed.py certifies (it exits 1 if the certified r' falls below the tabulated value, which is passed as a target; all 18 pass, run_lowseeds.sh), on every piece of $(0, r']$ with r' as in Table 1:*

- $|Z_N^s/\text{Main}^s - 1| \leq B_s$;
- $c_{\text{seed}}^s = (1 - B_s)\sqrt{\pi/2A}|S^s|$ and $\kappa_{\text{seed}} = q \max(0, -\text{Re } F(t_0))$;
- an enclosure $\omega \in \frac{S^-}{u_0 S^+} \cdot \frac{1+D(B_-)}{1+D(B_+)}$, and a lower bound for $\min_i |G_i|$ on it.

These hold for every $N \geq 151$ and $x = a/N$ in the piece, under the Δ -hypotheses of P1d Theorem 9, which hold by P1e Lemma H(4).

Table 1: Low seeds, twisted certificate on $(0, r']$ (raw output `lowseeds_pm.txt`; lower bounds rounded down, upper bounds up).

seed	r'	$\max B_{\pm} \leq$	$\min c_{\text{seed}}^{\pm} \geq$	$\kappa_{\text{seed}} \leq$	margin $\geq$	$ \omega \leq$
1/2	0.002075	0.0572	0.7866	6.45e-5	1.336	4.787
1/3	0.001	0.0386	0.7708	0.00128	0.4902	2.918
2/3	0.00075	0.0240	0.7823	0	0.4999	2.767
1/4	0.00075	0.0303	0.6914	6.01e-8	0.2132	1.825
3/4	0.000175	0.0104	0.7061	1.42e-8	0.2794	1.678
1/5	0.00025	0.0631	0.5657	0	0.1842	1.253
2/5	0.003	0.103	0.7589	4.02e-5	0.1220	4.519
3/5	0.0015	0.0332	0.8128	0	0.3026	3.570
2/7	0.001	0.0429	0.7344	0	0.4195	2.534
5/7	0.001	0.0428	0.7329	7.95e-6	0.4104	2.516
3/7	0.002	0.0588	0.8173	1.67e-6	0.5592	4.256
4/7	0.002	0.0573	0.8180	0	0.5351	4.249
3/8	0.002	0.0365	0.7949	6.32e-11	0.2043	3.310
5/8	0.002	0.0387	0.7923	6.32e-11	0.2100	3.322
2/9	0.00025	0.0636	0.6187	3.61e-6	0.1349	1.587
7/9	0.00025	0.0639	0.6173	0	0.1337	1.582
4/9	0.0015	0.0481	0.8090	7.77e-8	0.7893	4.392
5/9	0.002	0.0722	0.7884	0	0.6703	4.652

The radius r' is the largest piece edge up to which every piece has $B_{\pm} < 1$ and margin > 0.05 . Beyond r' the certified B_- reaches 1.46 (seed 3/4), so a twisted analogue of P1e's full-radius low-seed certificate fails there. Proposition 12 takes over. At the seed 1/2 the ω -limit is finite and far from the excluded set: $Z_2^{\pm} = e^{\Phi_2(1)}$, so $\omega_2 = 1/u_2$ and $t_1^* \approx \frac{3}{2}$. Near $\zeta = -1$ this is the reason the normalised G_i (not the P1f factors) are used.

Proposition 12 ($A^{\pm}$: refined direct covering; PROVED, computer-assisted). *Let $\mathcal{D}$ consist of the following discs:*

- the hazard discs of P1e for $10 \leq n \leq 59$;
- the low-seed discs of radius $0.999r'$, with left radius $0.999r'((q-p)/q)$ and right radius $0.999r'(p/q)$ (file `caseA_discs.txt`).

For every $x \in [\frac{1}{5}, \frac{1}{2}] \setminus \mathcal{D}$ that lies in no hazard of height ≥ 60 , and every $N \geq 151$ with $x = a/N$, $\gcd(a, N) = 1$:

$$|Z_N^{\pm}| \geq 0.4799, \quad \min_i |G_i| \geq 0.05011, \quad |\omega| \leq 7.954.$$

Proof. For $M < 60 \leq N$, split $Z_N^s = P_M^s(x) + \mathcal{R}$, where $P_M^s(x) = \sum_{n \leq M} \psi_n(x) e(-(n^2 + sn)x)$.

The polynomial part. For $n \leq M$, ψ_n depends only on $a_m = u_0^m / (m(1 - e(-mx)))$, $m \leq M$ (no $m \leq M$ is a multiple of N). The only N -dependence is through $u_0 \in \frac{1}{c}(1 + D(0.1e^{-151\ell}))$ (P1d Lemma W). So P_M^s is enclosed on an x -interval uniformly in $N \geq 151$.

The tail. e^{Φ} is coefficientwise majorised by $\exp(\sum_{m < 60} \alpha_m Y^m + \sum_{m \geq 60} |a_m| Y^m)$ with $\alpha_m \geq \sup |a_m|$. By P1e Lemma H(3), $\sum_{m \geq 60} |a_m| \leq \tau/59$. The disc hypothesis of H(3) is met: a low rational i'/m' within $y^{60}/(4\tau) = 1.1 \cdot 10^{-17}$ of x would put x in its disc of radius $0.999r' \geq 1.7 \cdot 10^{-4}$, and every low rational in $[\frac{1}{5}, \frac{1}{2}]$ has a disc. The second exponent has only degrees $\geq 60 > M$. Therefore

$$\sum_{n > M} |\psi_n| \leq T_M := e^{\sum_{m < 60} \alpha_m + \tau/59} - \sum_{n \leq M} c_n,$$

where the c_n are the coefficients of $\exp(\sum_{m < 60} \alpha_m Y^m)$. (Correction, Reviewer AC: the script previously excluded the hazard discs of height $53 \leq n \leq 59$ with radius $25 \cdot 2^{-n}$, twice $y^n/(4\tau) = 12.5 \cdot 2^{-n}$, so the annuli $12.5 \cdot 2^{-n} \leq |x - j/n| < 25 \cdot 2^{-n}$ were not covered. Fixed to $25 \cdot 2^{-(n+1)}$ and all four blocks rerun on 2026-09-24; the interval count and all constants below are unchanged.) `P1g_caseA.py` covers $[\frac{1}{5}, \frac{1}{2}] \setminus \mathcal{D}$ adaptively ($M \in \{8, 20, 40, 59\}$, bisection), in four blocks, with 12,205 intervals and 0 failures (`caseA.out.txt`). On each interval it encloses $Z^\pm \in P_M^\pm + D(T_M)$, ω and the G_i in Arb, and checks $|Z^\pm| > 0.05$ and $\min |G_i| > 0.05$. A failure is fatal (exit 1). This was tested with the threshold 0.3, above the true minimum 0.232: the script failed at $x = 0.22$. $\square$

Points inside hazards of height ≥ 60 are covered by the intervals but not by the proposition. They are handled by the hazard step below, exactly as in P1e.

Lemma 13 ($B^\pm$; PROVED, computer-assisted, tail by hand). *Let j/n be a hazard seed, $0 < |d| < \varepsilon_n$ (P1e hazard radii), and assume the Δ -hypothesis of P1e Lemma H(2).*

- (i) Exact seeds $10 \leq n \leq 150$ (all 4098 in $[\frac{1}{5}, \frac{4}{5}]$): with the level- n data of Proposition 10, the exact sums of P1e Lemma B' (extended to $53 \leq n \leq 150$ with the radius $y^n/(4\tau)$), and Lemma 7 in crude form, we get

$$E_\pm \leq 0.0048, \quad R_\pm \leq 0.0040, \quad \rho \leq 0.0361, \quad \min_i |G_i|(j/n) - \sigma \geq 0.2208$$

(`P1g_lemmaB.py exact`; per- n maxima in `lemmaB_exact.txt`).

- (ii) Deep seeds $n \geq 151$: under the induction hypothesis $|Z_n^\pm(j/n)| \geq c_0 e^{-\kappa n}$ ($c_0 = 0.02$, $\kappa = 0.005$) and $|\omega_n| \leq 10$, the uniform region bounds of P1e (with g_i enlarged by the factor $1 + 10^{-6}$) give

$$E_\pm, R_\pm \leq 1.9 \cdot 10^{-14}, \quad \sum_{151 \leq n \leq 6000} \sigma(n) \leq 4.96 \cdot 10^{-12}, \quad \sum_{151 \leq n \leq 6000} \rho(n) \leq 3.31 \cdot 10^{-13}$$

(`P1g_lemmaB.py crude 151 6000 10`), with all side conditions checked, including the new one, $v_0 \geq 2\pi d/A'$.

- (iii) Tail $n > 6000$ (by hand, not machine-checked; as in P1e). Here $\bar{w}, R_0 \leq 2^{-n}$, $d \leq 12.5 \cdot 2^{-n}$ and $n_2 \geq 2^n/(25n) - 1$. Also $\beta e^{\pi n} \leq 1.4 \cdot 2^{n/2}$, $P_1 + \pi \leq 4n$ and $P_2 \leq 40n$. So $r_1 + r_3 \leq (400n^2 + 3n + 110)2^{-n/2}$, while $r_2, r_4, r_5 \leq \exp(-10^{-5}2^n)$. This gives $E \leq 10^5 n^{5/2} (e^{0.005}/\sqrt{2})^n < 10^{-860}$, and $R \leq 10^4 n^{5/2} (e^{0.005}/\sqrt{2})^n < 10^{-860}$ as in P1e. Further, $\epsilon_u \leq 160 \cdot 2^{-n}$. Put $\theta := e^{0.005}/\sqrt{2} = 0.71066$. Since $E, R < 10^{-860}$, $\frac{(1+R)(1+E)}{(1-R)(1-E)} \leq 1 + 2.001(E+R)$, so by Lemma 9

$$\rho \leq 2\epsilon_u + 4.02(E+R) \leq 320 \cdot 2^{-n} + 4.43 \cdot 10^5 n^{5/2} \theta^n \leq 4.5 \cdot 10^5 n^{5/2} \theta^n,$$

using $2^{-n} \leq 10^{-600} n^{5/2} \theta^n$ for $n > 6000$ (as $2\theta > 1.42$). With $|\omega_q| \leq 10$ and $|d| \leq 12.5 \cdot 2^{-n}$,

$$\sigma(n) \leq 15\rho + 12.5 \cdot 2^{-n} \left(\frac{33\pi}{2} + 4\pi \right) \leq 7 \cdot 10^6 n^{5/2} \theta^n.$$

The ratio of consecutive terms is $\leq \theta(1 + \frac{1}{6000})^{5/2} < 0.7110$, so $\sum_{n > 6000} \sigma(n) \leq 7 \cdot 10^6 \cdot 6001^{5/2} \theta^{6001} / 0.289 < 10^{-870}$, and likewise $\sum_{n > 6000} \rho(n) < 10^{-871}$. (Correction, Reviewer AC: an earlier version stated $\rho \leq 500 \cdot 2^{-n}$ and $\sigma(n) \leq 9000 \cdot 2^{-n}$, which ignored the E, R contributions to ρ ; those decay only like $n^{5/2} \theta^n$. The conclusion $S \leq 4.97 \cdot 10^{-12}$ is unaffected.)

The crude hazard discs are 10^{-15} wide or less, and every hazard seed j/n ($n \geq 10$) is at distance $\geq 1/(qn) \gg \varepsilon_n$ from each region boundary $\frac{1}{5}, \frac{1}{4}, \frac{1}{3}, \frac{2}{5}, \frac{1}{2}$. So a hazard disc never crosses a region boundary. The factor $1 + 10^{-6}$ is extra safety for $|u_0|$ on the disc.

4 The induction

Theorem 14 (main; PROVED, computer-assisted). *For all $N \geq 3$ and $\gcd(a, N) = 1$ with $\frac{a}{N} \in [\frac{1}{5}, \frac{4}{5}] \setminus \{\frac{1}{2}\}$:*

($M^\pm$) $|Z_N^\pm(a/N)| \geq c_0 e^{-\kappa N}$, with $c_0 = 0.02$ and $\kappa = 0.005$;

(Ω) $\min_i |G_i| \geq 0.05 - S$ and $|\omega| \leq 7.954 \prod_{n \geq 151} (1 + \rho(n)) \leq 7.954(1 + 10^{-12})$, where $S := \sum_{n \geq 151} \sigma(n) \leq 4.97 \cdot 10^{-12}$. Hence $\min_i |G_i| \geq 0.0499$ and $|\omega| \leq 7.96$.

Proof. By Lemma 4, take $x = a/N \in [\frac{1}{5}, \frac{1}{2})$. We use strong induction on N , with the hypothesis strengthened to $\min_i |G_i| \geq 0.05 - S(N)$ and $|\omega| \leq 7.954 \prod_{151 \leq m < N} (1 + \rho(m))$, where $S(N) := \sum_{151 \leq m < N} \sigma(m)$ and $\rho(m), \sigma(m)$ are the crude bounds of Lemma 13(ii),(iii). Since $\prod (1 + \rho) \leq 1 + 10^{-12}$, the $|\omega|$ -hypothesis implies $|\omega| \leq 10$, which is the value used in Lemma 13(ii).

Base $N \leq 150$: Proposition 10, which gives $|Z^\pm| \geq 0.6067$, margin ≥ 0.2331 and $|\omega| \leq 4.236$.

Step $N \geq 151$. The case split of P1e Theorem M, with the low-seed discs shrunk to r' :

1. x lies in a hazard with $n \geq 151$; take the deepest one, j/n . P1e Lemma H(2) gives the Δ -hypothesis. The induction hypothesis at level $n < N$ gives $|Z_n^\pm| \geq c_0 e^{-\kappa n}$, margin $\geq 0.05 - S(n)$ and $|\omega_n| \leq 10$. By Lemma 13(ii) and Lemma 8:

$$|Z_N^\pm| \geq (1 - E) \sqrt{\pi/2A} e^{-\pi R_0} (1 - R) |T^\pm| e^{-n|F(t_0)|N} \geq c_0 e^{-\kappa N},$$

as in P1e: $N \geq 1/(n|d|)$ and $n|F(t_0)| \leq g^n/(5n) + n\pi R_0^2 \leq 2^{-n}$. By Lemma 9, the margin is $\geq 0.05 - S(n) - \sigma(n) \geq 0.05 - S(N)$, and $|\omega_N| \leq |\omega_n|(1 + \rho(n))$.

2. The deepest hazard has $10 \leq n \leq 150$. The same argument with Lemma 13(i) and the exact level- n data: $|Z_N^\pm| \geq \frac{1}{2} \cdot 0.7 \cdot 0.99 \cdot 0.6067 e^{-2^{-10}N} \geq c_0 e^{-\kappa N}$. The margin is ≥ 0.2208 , and $|\omega_N| \leq 4.236 \cdot 1.037$.
3. x lies in no hazard with $n \geq 10$, but $0 < |x - p/q| \leq r'$ for a low seed. Proposition 11: $|Z_N^\pm| \geq c_{\text{seed}} e^{-\kappa_{\text{seed}} N} \geq 0.5657 e^{-0.00128N}$, margin ≥ 0.122 and $|\omega| \leq 4.787$.
4. Otherwise. Proposition 12: $|Z_N^\pm| \geq 0.4799$, margin ≥ 0.05011 and $|\omega| \leq 7.954$.

The cases are exhaustive: case 4 is the complement of the hazards of height ≥ 10 and of the r' -discs. Only case 1 lowers the margin below its terminal value, and by at most $\sigma(n)$ with $n \geq 151$ each time. $\square$

Corollary 15 (PROVED, computer-assisted; standing `thm:model`, `thm:RJ`). *For every $N \geq 16$ and every $\zeta = e(a/N)$ with $\frac{1}{5} \leq \frac{a}{N} \leq \frac{4}{5}$ ($\zeta \neq -1$ automatically), the hypotheses of P1f Theorem C hold: $Z_N \neq 0$ (P1e Theorem M), $Z_N^+ \neq 0$, and ω avoids the six values (Lemma 2). Hence:*

- the poles of U and of V accumulate at $\pm\sqrt{\zeta}$ and $\pm\sqrt{\bar{\zeta}}$;
- the accumulation set of these poles on $|x| = 1$ contains the closed arcs $\arg x \in [\frac{\pi}{5}, \frac{4\pi}{5}] \cup [\frac{6\pi}{5}, \frac{9\pi}{5}]$.

Proof. P1f Theorem C(i)–(iii) with Theorem 14, and P1e Theorem M for $Z_N \neq 0$ on $[\frac{1}{5}, \frac{4}{5}]$ (reference added at the request of Reviewer AC). For the arcs: the accumulation set is closed and contains $\pm e(\frac{a}{2N})$ and $\pm e(-\frac{a}{2N})$ for all admissible a/N , $N \geq 16$, and these points are dense in the two arcs. $\square$

5 Numerical cross-checks (VERIFIED)

- `P1g_verify.py`: the direct ω_N (exact finite formula, floating) against the certified model $S^-(u_0S^+)$ of Lemma 6.
 - At 1/3: $|\Delta\omega| = 7.1 \cdot 10^{-3}$ at $d = 1.5 \cdot 10^{-3}$, and $2.4 \cdot 10^{-2}$ at $d = 3.7 \cdot 10^{-3}$.
 - At 3/4: $7.4 \cdot 10^{-4}$ at $d = 8.3 \cdot 10^{-5}$.
 - At 2/5: $8.8 \cdot 10^{-3}$ at $d = 6.7 \cdot 10^{-4}$.

All are $O(d)$ and inside the certified $|\omega|(B_+ + B_-)$. The refined direct enclosures of Proposition 12 at 71/301, 97/400, 133/401, 180/499 contain the direct values.

- `P1g_margin_profile.py`: sharp point enclosures (tail $T_M \leq 2 \cdot 10^{-3}$) at $x = k/20000$, 5923 points. The $N \rightarrow \infty$ margin profile has its minimum 0.2319 at $x \approx 0.227$, a smooth minimum of the factor G_3^- . It matches the base minimum 0.2331 at 22/97. So the certified threshold 0.05 is conservative. Its near-attainment (0.05011) comes from wide intervals at hazard and disc edges, where the bisection stops as soon as the enclosure clears 0.05.

6 What is not proved; weakest points

1. **Adaptations (reviewed).** Lemma 7 is a transcription of P1d Lemma 8 and P1e Lemma O with a bounded analytic factor inserted. The only genuinely new estimate is the lower-ray tail (halved Gaussian). Both scripts are line-by-line edits of the reviewed `P1e_cert.py` and `P1e_lemmaB.py`, and were checked by Reviewer AC; the gaps found are corrected in place above and all certificates were rerun.
2. **Hand-checked tail** (Lemma 13(iii), $n > 6000$). The constants are generous, but the tail is not machine-checked, as in P1e Lemma B for $n > 3000$.
3. **Inherited standing.** Theorem 14 rests on P1d (Lemmas W, 6, 8, Theorems 2, 5, 9), P1e (Lemmas H, R, O and the hazard scheme; reviewed CORRECT by Reviewer AA), and `python-flint/Arb`. Corollary 15 further rests on P1f Theorem C, whose Laplace $O(t)$ and index j_0 are not explicit, and on the standing Theorems `thm:model`, `thm:RJ`.
4. **Precision floor.** `P1g_lemmaB.py` runs at 200 bits. For $n \gtrsim 400$ the printed $\rho(n)$ sits at the rounding floor $1.6 \cdot 10^{-59}$. These are valid upper bounds, merely not sharp, and their sum over $n \leq 6000$ is included in the $4.96 \cdot 10^{-12}$.
5. $N \leq 15$. Theorem 14 holds there, but P1f Theorem B (zeros of B) is proved only for $N \geq 16$. So Corollary 15 is stated for $N \geq 16$; this does not affect density.
6. **Scope.** Only the middle arc $[\frac{1}{5}, \frac{4}{5}]$ is treated. The edge $\ell < 0.855$ is OPEN, as in P1e.

Files (in `rootunity_zeros/general/P1g/`):

- `P1g_seeds.py`: Proposition 10; data in `seeds_*.txt`.
- `P1g_cert.py`: Lemma 7 at the low seeds.
- `P1g_lowseed.py` and `run_lowseeds.sh`: Proposition 11; output in `lowseeds_pm.txt`.
- `P1g_caseA.py`: Proposition 12; `caseA_discs.txt`, `caseA_out.txt`.

- `P1g_lemmaB.py`: Lemma 13; `lemmaB_exact.txt`, `lemmaB_crude_151_6000.txt`.
- `P1g_inherit_check.py`, `P1g_verify.py`, `P1g_margin_profile.py`, `P1g_scope.py`: VERIFIED checks only.
- `P1g_certlog.txt`: raw outputs.

Every run used `perl -e 'alarm 290; exec @ARGV'`. All runs were single python-flint processes under 300 MB (fresh check: 24 GB RAM). No Rust was needed, because no enumeration beyond these Arb certificates was required.